

# A Technical Comment on Claims of Infinite Fatigue Life and Self-Healing Dislocation Networks in Hierarchical Alloys

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## ABSTRACT

We provide a technical comment on the recent preprint by Kawabata et al. reporting complete suppression of crack nucleation up to  $10^{12}$  cycles in a hierarchically structured austenitic steel. The original work contains a detailed processing description, a plausible qualitative mechanism, and internally consistent mechanical data. At the same time, several of its claims conflict with the standard framework of high-cycle fatigue, notably the existence of a non-zero lower bound on damage accumulation rate set by thermally activated processes. We enumerate the specific claims that require independent verification, identify measurement resolution limits that could mimic apparent infinite life, and propose a minimal set of falsifiable predictions that a supporting theory should satisfy. We argue that enthusiasm for the reported behavior should be tempered with structured skepticism pending replication.

## 1 Introduction

The recent preprint by Kawabata and co-workers [1] has generated substantial interest inside and outside the materials community for reporting a hierarchical metallic architecture that appears to exhibit no fatigue damage after  $10^{12}$  cycles. A concurrent letter by Zhang et al. [2] reports qualitatively similar behavior in a different alloy family.

If correct, these results would overturn the quantitative framework of high-cycle fatigue that has been built up over more than eight decades of metallurgical research, and they would have immediate consequences for aerospace certification, rotating machinery design, and civil infrastructure. The magnitude of the implied claims makes a careful, structured response essential.

The purpose of this Comment is to (i) identify the specific experimental claims that carry the greatest theoretical burden, (ii) note measurement resolution issues that could produce the appearance of infinite life, and (iii) propose a minimal set of predictions against which a supporting theory could be falsified.

## 2 What exactly is claimed

We distinguish three claims of decreasing strength:

- **C1. Strength-ductility.** The alloy exhibits yield stress 1.64 GPa (238 ksi) and uniform elongation 37%. This claim is strong but not unprecedented; similar combinations have been approached in nanotwinned and heterogeneous structures.
- **C2. Extended endurance.** No measurable stiffness

degradation up to  $10^{12}$  cycles at  $\sigma_a/\sigma_y \approx 1.0$ . This is one order of magnitude beyond the largest fatigue-cycle counts reported in the peer-reviewed literature for conventional alloys.

- **C3. Mechanistic absence of persistent slip bands.** Post-cycling TEM does not show the characteristic extrusion-intrusion morphology. This claim is the most theoretically consequential, as it would imply that the fundamental unit of fatigue damage in fcc metals is suppressible by structure alone.

C1 is plausible. C2 is extraordinary. C3 is the mechanistic linchpin; without it, C2 is phenomenology without explanation.

## 3 Resolution limits that could mimic infinite life

The fatigue limit in the original work is inferred from the absence of detectable change in dynamic modulus and first-mode resonance frequency. Both observables have finite resolution. A back-of-envelope calculation indicates that a damage rate  $dD/dN < 10^{-13}$  per cycle, which is below the reported detection floor, would still produce complete failure within  $10^{14}$  cycles.

More importantly, classical fatigue proceeds through stages that are not uniformly detectable by bulk elastic observables. Subsurface microvoid nucleation in stage I, which is characteristic of long-life fatigue in clean alloys, is often invisible to bulk stiffness measurements until the final 5 to 10% of life. Reported invariance of the resonance spectrum therefore does not rule out ongoing damage accumulation.

Recommended additional measurements include (i) high-resolution X-ray diffraction peak broadening with long counting time, (ii) thermoelastic stress analysis at the specimen surface, (iii) positron annihilation spectroscopy before and after long-duration cycling, and (iv) destructive metallography on interrupted specimens at logarithmically spaced cycle counts.

#### 4 Falsifiable predictions

A theory that genuinely supports C2 and C3 should make predictions beyond the existence proof provided by the data. We propose the following minimum set:

- *P1. Scale coupling constant.* The model of Kawabata et al. introduces a parameter  $\gamma D_{ij}$  coupling scale  $i$  to scale  $j$ . The theory should predict the frequency dependence of the damage rate with no adjustable parameters beyond those fit from monotonic data.
- *P2. Temperature window.* Thermally activated detrapping of dislocations from the hierarchical network should eventually dominate at elevated temperature. The theory should predict the temperature  $T^*$  at which apparent infinite life breaks down.
- *P3. Composition transferability.* If the mechanism is geometric rather than chemistry specific, it should be reproducible across at least three chemically distinct alloy systems with similar hierarchical geometry.

The concurrent observation by Zhang et al. [2] is consistent with P3 at the level of two alloy families and provides preliminary support.

#### 5 Position

We take the position that the reported results are interesting, technically serious, and sufficient to justify substantial replication effort at a small number of well-instrumented laboratories, but that they are not sufficient to justify design changes in service-critical applications, nor do they currently warrant revision of pedagogical treatments of fatigue. The historical reference class for extraordinary materials claims, including the recent LK-99 episode [3], is that independent replication is the single most important filter. We recommend that the community wait for at least three independent replications on chemically distinct alloys before drawing strong conclusions.

#### References

- [1] Y. Kawabata et al., *arXiv* 2604.11891 (2026).
- [2] L. Zhang et al., *arXiv* 2604.12207 (2026).
- [3] S. Lee, J. Kim, Y.W. Kwon, *arXiv* 2307.12008 (2023).